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Inventor(s)

PARK; Eun Suk et al.

SECONDARY BATTERY AND METHOD OF MANUFACTURING THE SAME

Abstract

A secondary battery includes: an electrode assembly; a first exterior material enclosing the electrode assembly such that the electrode assembly is sealed; a second exterior material enclosing an outer surface of the first exterior material, wherein at least a portion of the second exterior material is unbonded to the first exterior material; and an electrode lead electrically connected to the electrode assembly and configured to extend from an interior to an exterior of the first exterior material and the second exterior material.

Inventors: PARK; Eun Suk (Daejeon, KR), HWANG; Ji Young (Daejeon, KR), LEE; Yu Jin (Daejeon, KR), JU; Hye Yeong (Daejeon, KR), KIM; Sang Hun (Daejeon, KR), YU; Hyung Kyun (Daejeon, KR)

Applicant: LG ENERGY SOLUTION, LTD. (Seoul, KR)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is based on and claims priority from Korean Patent Application No. 10-2024-0029900, filed on Feb. 29, 2024, with the Korean Intellectual Property Office, the disclosure of which is incorporated herein in its entirety by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to a rechargeable secondary battery and a method of manufacturing the same.

BACKGROUND

[0003] In recent years, with the depletion of fossil fuels leading to rising energy source prices and growing concerns about environmental pollution, the demand for eco-friendly alternative energy sources has become an essential factor for future living. Accordingly, research on various power generation technologies, such as solar power, wind power, and tidal power, has been continuously conducted, and there has also been significant interest in power storage devices, such as batteries, to use the generated electric energy more efficiently.

[0004] Furthermore, as the development of technology and demand for electronic mobile devices and electric vehicles using batteries grow, the demand for batteries as an energy source rapidly increases, leading to extensive research on batteries capable of meeting various requirements.

[0005] Batteries for storing electric energy are generally categorized into primary batteries and secondary batteries. Primary batteries are disposable and intended for single use, whereas secondary batteries are rechargeable and manufactured using materials capable of repeating oxidation and reduction processes between electric current and materials. In secondary batteries, when a reduction reaction occurs on a material due to electric current, the batteries are charged, and when an oxidation reaction occurs on the material, the batteries are discharged. When this charging and discharging is repeated, electricity is generated.

SUMMARY

[0006] The present disclosure provides a secondary battery with improved quality and performance by preventing or suppressing sealing defects and insulation defects, and a method for manufacturing a secondary battery with enhanced manufacturing process efficiency.

[0007] A secondary battery according to an embodiment of the present disclosure may include: an electrode assembly; a first exterior material enclosing the electrode assembly such that the electrode assembly is sealed; a second exterior material enclosing an outer surface of the first exterior material, wherein at least a portion of the second exterior material is unbonded to the first exterior material; and an electrode lead electrically connected to the electrode assembly and provided to extend from the interior to the exterior of the first exterior material and the second exterior material.

[0008] The first exterior material may have thermal sealability and gas permeability.

[0009] The secondary battery may further include an electrolyte accommodated within the interior of the first exterior material along with the electrode assembly.

[0010] The second exterior material may have thermal sealability, and may include a sealant layer provided to face the first exterior material.

[0011] At least a portion of the sealant layer may be bonded to the first exterior material by sealing.

[0012] The second exterior material may be bonded to the sealant layer provided to face the first

exterior material by sealing such that the first exterior material is sealed.

[0013] The second exterior material may include a metal layer including a metallic material, and an insulating layer including an insulating material.

[0014] The second exterior material may include a metal layer including stainless steel (SUS), and an outer surface of the metal layer may be coated with nylon.

[0015] The second exterior material may include at least one hole formed in a portion thereof.

[0016] A secondary battery manufacturing method according to an embodiment of the disclosure may include: sealing a first exterior material accommodating an electrode assembly in a cup portion; charging and discharging the electrode assembly provided in the sealed first exterior material to activate the electrode assembly; discharging gas collected in a gas collection portion of the first exterior material through the activation; removing the gas collection portion of the first exterior material; and enclosing the first exterior material with a second exterior material.

[0017] The secondary battery manufacturing method may further include sealing an area between the cup portion and the gas collection portion after the discharging the collected gas.

[0018] The secondary battery manufacturing method may further include forming at least one hole in a portion of the second exterior material before the enclosing the first exterior material.

[0019] In the enclosing the first exterior material, the first exterior material and the second exterior material may be bonded to each other by thermal sealing.

[0020] A battery pack according to another aspect of the present disclosure may include the secondary battery according to an aspect of the present disclosure.

[0021] A vehicle according to yet another aspect of the present disclosure may include the battery pack according to the another aspect of the present disclosure.

[0022] According to an embodiment of the present disclosure, a secondary battery may include a first exterior material and a second exterior material.

[0023] Accordingly, performance degradation of the secondary battery due to sealing defects may be prevented or suppressed.

[0024] In addition, wrinkles on the second exterior material caused by gas generated during the activation process may be prevented or suppressed.

[0025] As a result, issues such as insulation defects in the secondary battery may be reduced.

[0026] According to an embodiment of the present disclosure, the secondary battery may have a hole formed in the second exterior material.

[0027] Accordingly, gas generated inside the secondary battery may be efficiently discharged to the exterior.

[0028] Through this, functional degradation of the secondary battery may be prevented or suppressed.

[0029] In addition, the occurrence of the venting phenomenon of the pouch caused by gas inside the exterior material may be delayed or prevented, thereby improving the safety of the secondary battery.

[0030] According to an embodiment of the present disclosure, in the secondary battery manufacturing method, the second exterior material may be bonded after the activation process and the degassing process are completed.

[0031] Accordingly, the cost consumed for the removal of a gas collection portion may be reduced, improving the economic efficiency of the process.

[0032] Furthermore, the degree of flexibility in hole formation may increase, and bonding of the second exterior material may be facilitated, enhancing the efficiency of the process.

[0033] The effects according to the present disclosure are not limited to the features exemplified above, and more diverse effects are included in this specification.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0034] The following drawings attached hereto illustrate example embodiments of the present disclosure and, together with the detailed description to be described later, serve to further understand the technical idea of the present disclosure. Therefore, the present disclosure should not be construed as being limited to the matters illustrated in the drawings.

[0035] FIG. 1 is a perspective view schematically illustrating a secondary battery according to Embodiment 1 of the present disclosure.

[0036] FIG. 2 is a partially cutaway perspective view schematically illustrating the secondary battery according to Embodiment 1 of the present disclosure.

[0037] FIG. 3 is a cross-sectional view schematically illustrating a cross-section taken along A-A' of FIG. 1.

[0038] FIG. 4 is a perspective view schematically illustrating a secondary battery according to Embodiment 2 of the present disclosure.

[0039] FIG. 5 is a flowchart schematically illustrating a secondary battery manufacturing method according to Embodiment 3 of the present disclosure.

[0040] FIG. 6 is a plan view schematically illustrating a state where the first exterior material encloses the electrode assembly in the secondary battery manufacturing method according to Embodiment 3 of the present disclosure.

[0041] FIG. 7 is a plan view schematically illustrating a state where a gas collection portion is removed in the secondary battery manufacturing method according to Embodiment 3 of the present disclosure.

[0042] FIG. 8 is a plan view schematically illustrating a state where the second exterior material encloses the first exterior material in the secondary battery manufacturing method according to Embodiment 3 of the present disclosure.

[0043] FIG. 9 is a perspective view illustrating an automobile including a battery pack according to an embodiment of the present disclosure.

[0044] In some of the attached drawings, corresponding components are given the same reference numerals. Those skilled in the art would appreciate that the drawings depict elements simply and clearly and have not necessarily been drawn to scale. For example, in order to facilitate understanding of various embodiments, the dimensions of some elements illustrated in the drawings may be exaggerated compared to other elements. Additionally, elements of the known art that are useful or essential in commercially viable embodiments may often not be depicted so as not to interfere with the spirit of the various embodiments of the present disclosure.

DETAILED DESCRIPTION

[0045] Hereinafter, with reference to the accompanying drawings, embodiments of the present disclosure will be described to enable those ordinarily skilled in the art to which the present disclosure pertains to easily implement the embodiments. However, the present disclosure may be implemented in various different forms and is not limited or restricted by the following examples.

[0046] In order to clearly describe the present disclosure, detailed descriptions of portions unrelated to the present disclosure or known technologies that may unnecessarily obscure the gist of the present disclosure have been omitted. Additionally, when assigning reference numerals to components in each drawing in this specification, the same or similar reference numerals are assigned to the same or similar components throughout the specification.

[0047] In addition, the terms or words used in the specification and claims should not be construed as limited to their ordinary or dictionary meanings, but should be construed as meanings and concepts consistent with the technical idea of the present disclosure based on the principle that an inventor may appropriately define the concepts of terms in order to explain their invention in the best way.

[0048] The secondary battery may be classified into, for example, a cylindrical cell, a pouch cell,

and a prismatic cell, depending on the shape. Among these, the pouch cell accommodates an electrode assembly, in which a positive electrode, a negative electrode, and a separator are stacked inside a pouch, and the outer portion of the pouch may be sealed during manufacturing.

[0049] A pouch cell manufacturing process may include a step of discharging gas collected inside a gas collection portion of the pouch, followed by a step of removing the gas collection portion. In addition, the pouch cell manufacturing process may include a step of resealing the pouch from which the gas collection portion has been removed. During this process, an electrolytic solution may adhere to the area cut for removing the gas collection portion. The electrolytic solution remaining on the cut area may cause poor sealing when resealing the pouch.

[0050] The present disclosure provides a secondary battery with improved quality and performance by preventing or suppressing issues of reduced sealability, as well as a manufacturing method for a secondary battery with enhanced process efficiency.

Embodiment 1

[0051] FIG. 1 is a perspective view schematically illustrating a secondary battery **10** according to Embodiment 1 of the present disclosure. FIG. 2 is a partially cutaway perspective view of the secondary battery **10** according to Embodiment 1 of the present disclosure, and FIG. 3 is a cross-sectional view schematically illustrating a cross-section taken along A-A' of FIG. 1.

[0052] The secondary battery **10** according to Embodiment 1 of the present disclosure may refer to a rechargeable secondary battery. For example, the secondary battery **10** may have a configuration in which an electrode assembly **100** is accommodated inside a case in the form of exterior material.

[0053] The electrode assembly **100** may include a positive electrode, a negative electrode, and a separator. Here, the separator may be disposed between the positive and negative electrodes to physically separate the negative and positive electrodes. The electrode assembly **100** may be in a stacked configuration in which the positive electrode, the negative electrode, and the separator are stacked, or in a jelly-roll configuration in which the positive electrode, the negative electrode, and the separator are wound.

[0054] Conventional secondary batteries, in which an electrode assembly is disposed inside a pouch, have a problem in that sealing defects are likely to occur during the manufacturing process, leading to a decrease in the performance of the secondary battery. Conventional pouch-type secondary batteries may include a process of removing the space where gas generated during the activation process is collected and resealing the pouch. In the manufacturing process of conventional pouch-type secondary batteries, sealing defects may occur due to the electrolyte solution remaining on the sealing portions of the pouch, and these sealing defects may lead to insulation defects. Insulation defects may cause a decrease in the performance of the completed secondary batteries.

[0055] As an example of a configuration to prevent or suppress problems such as insulation defects, a secondary battery **10** according to Embodiment 1 of the present disclosure may include a first exterior material **200** and a second exterior material **300** for sealing an electrode assembly **100**.

[0056] Referring to FIGS. 1 and 2, the secondary battery **10** according to Embodiment 1 of the present disclosure may have a configuration in which the first exterior material **200** and the second exterior material **300** accommodate the electrode assembly **100**. The first exterior material **200** may enclose the electrode assembly **100** such that the electrode assembly **100** is sealed. In addition, the second exterior material **300** may enclose the outer surface of the first exterior material **200**.

[0057] For example, the first exterior material **200** may primarily seal the electrode assembly **100**, and the second exterior material **300** may secondarily seal the electrode assembly **100**. Unlike conventional pouches having multiple layers, the first exterior material **200** and the second exterior material **300** of the secondary battery **10** according to Embodiment 1 of the present disclosure may have separate configurations. Thus, the first exterior material **200** and the second exterior material **300** may be disposed such that at least some portions thereof have a gap therebetween. For example, in the first exterior material **200** and the second exterior material **300**, the portions that are

not sealed for sealing the electrode assembly **100** may not be bonded. In other words, the first exterior material **200** and the second exterior material **300** may be at least partially unbonded.

[0058] The interior of the first exterior material **200**, which primarily seals the electrode assembly **100**, may accommodate the electrolyte. For example, the electrolyte may be an electrolytic solution. Since the first exterior material **200** seals the electrode assembly **100** and the electrolytic solution in an accommodated state, the second exterior material **300** may prevent or suppress a decrease in sealability caused by the electrolytic solution. Accordingly, the secondary battery **10** according to Embodiment 1 of the present disclosure may prevent or suppress a decrease in performance caused by sealing defects. As a result, the secondary battery **10** may reduce the occurrence of problems such as insulation defects. Furthermore, the secondary battery **10** may improve sealing performance for the electrode assembly **100** and the electrolytic solution through double sealing provided by the first exterior material **200** and the second exterior material **300**.

[0059] Meanwhile, the secondary battery **10** may include an electrode lead **400**. The electrode lead **400** may be electrically connected to the electrode assembly **100**. The electrode lead **400** may be provided to extend from the interior of the first exterior material **200** and the second exterior material **300** to the exterior. Through the electrode lead **400** protruding outside the first exterior material **200** and the second exterior material **300**, the secondary battery **10** may supply electrical energy to the exterior. Accordingly, the electrode lead **400** may be a conductor.

[0060] The secondary battery **10** may further include a lead film **500**. The lead film **500** may cover the electrode lead **400** to insulate the first exterior material **200** and the electrode lead **400** from each other. For example, the lead film **500** may be disposed on both sides of the electrode lead **400** to cover the electrode lead **400**. A pair of lead films **500** may be provided and respectively disposed on both sides of the electrode lead **400**.

[0061] The lead film **500** may include a thermally sealable material. Accordingly, the lead film **500** and the first exterior material **200** may be bonded through sealing.

[0062] The first exterior material **200** of the secondary battery **10** according to Embodiment 1 of the present disclosure may be thermally sealable. For example, the first exterior material **200** may be made of a thermally sealable material. Thus, the first exterior material **200** may be sealed to accommodate the electrode assembly **100** therein. In addition, the first exterior material **200** may be bonded to the lead film **500** through sealing.

[0063] In order to ensure thermal sealability, the first exterior material **200** may be made of one or more materials selected from polyethylene, polypropylene, polycarbonate, polyethylene terephthalate, polyvinyl chloride, acrylic polymers, polyacrylonitrile, polyimide, polyamide, cellulose, aramid, nylon, polyester, poly (p-phenylene benzobisoxazole), polyarylate, Teflon, and glass fiber. For example, the first exterior material **200** may be made of polyolefin resin such as polypropylene (PP) or polyethylene (PE).

[0064] The second exterior material **300** of the secondary battery **10** according to Embodiment 1 of the present disclosure may include multiple layers. For example, the second exterior material **300** may include a thermally sealable sealant layer **310**.

[0065] Referring to FIG. 3, the sealant layer **310** of the second exterior material **300** may be provided to face the first exterior material **200**. For example, the sealant layer **310** may be disposed as the innermost layer among the multiple layers included in the second exterior material **300**.

[0066] Since the first exterior material **200** and the second exterior material **300** are separate exterior materials, all portions may not be bonded to each other. Thus, some portions of the first exterior material **200** and the second exterior material **300** may be disposed to have a gap therebetween. For example, the sealant layer **310** of the second exterior material **300** and the first exterior material **200** may be disposed to partially have a gap therebetween or not be bonded in some areas. FIG. 3 is an enlarged view illustrating a portion where the first exterior material **200** and the sealant layer **310** are disposed to have a gap therebetween.

[0067] Meanwhile, some portions of the first exterior material **200** may be bonded to the sealant

layer **310** of the second exterior material **300** through sealing. For example, the portions of the first exterior material **200** that are sealed for sealing the electrode assembly **100** may be bonded to the sealant layer **310** of the second exterior material **300** through sealing. The first exterior material **200** and the sealant layer **310** may be sealed by heat and pressure.

[0068] In order to ensure thermal sealability, the sealant layer **310** of the second exterior material **300** may be made of one or more materials selected from polyethylene, polypropylene, polycarbonate, polyethylene terephthalate, polyvinyl chloride, acrylic polymers, polyacrylonitrile, polyimide, polyamide, cellulose, aramid, nylon, polyester, poly (p-phenylene benzobisoxazole), polyarylate, Teflon, and glass fiber. For example, the sealant layer **310** may be made of a polyolefin resin such as polypropylene (PP) or polyethylene (PE).

[0069] Referring to FIG. 3, the second exterior material **300** of the secondary battery **10** according to Embodiment 1 of the present disclosure may include a metal layer **320** and an insulating layer **330**.

[0070] The metal layer **320** of the second exterior material **300** may include a metallic material. For example, the metal layer **320** may include aluminum (Al). Since the second exterior material **300** includes the metal layer **320**, the structural rigidity of the secondary battery **10** may be improved.

[0071] The insulating layer **330** of the second exterior material **300** may include an insulating material. Referring to FIG. 3, the insulating layer **330** may be disposed as the outermost layer among the multiple layers included in the second exterior material **300**. For example, the insulating layer **330** may include a nylon-based resin. The insulating layer **330** may insulate the metal layer **320** from the exterior.

[0072] Meanwhile, as the secondary battery **10** undergoes repeated charging and discharging, gas may be generated, causing the internal pressure of the first exterior material **200** to increase. When the internal pressure of the first exterior material **200** increases excessively, for example, venting may occur, causing the secondary battery **10** to lose functionality.

[0073] For example, the interior of the first exterior material **200** of the secondary battery **10** may accommodate an electrolytic solution along with the electrode assembly **100**. At this time, residual moisture in the electrolytic solution inside the first exterior material **200** of the secondary battery **10** or moisture infiltrated from the outside may react with lithium salts to generate HF, and gases such as carbon dioxide, carbon monoxide, ethylene, and methane may be generated due to, for example, the decomposition of the electrolytic solution. In addition, depending on the material of the positive electrode included in the electrode assembly **100** of the secondary battery **10**, hydrogen and HF may be further generated, which may result in overheating due to overcharging and internal short circuits during charging and discharging. As a result, a large amount of gas may be generated inside the first exterior material **200**. Such gas may increase the internal pressure of the first exterior material **200**, and the increased pressure may cause swelling of the first exterior material **200** or venting, where the first exterior material **200** partially ruptures.

[0074] The secondary battery **10** according to Embodiment 1 of the present disclosure may include a structure capable of discharging internal gas to the exterior to prevent or suppress venting. For example, the first exterior material **200** may have gas permeability. For example, the first exterior material **200** may be made of a gas-permeable material. Thus, gas generated inside the first exterior material **200** may permeate through the first exterior material **200** and be discharged to the exterior of the first exterior material **200**.

[0075] The secondary battery **10** may include the first exterior material **200** and the second exterior material **300**. Accordingly, when the gas permeating through the first exterior material **200** is not discharged to the exterior of the second exterior material **300**, the second exterior material **300** may swell or rupture.

[0076] As an example of a configuration for discharging gas, the second exterior material **300** of the secondary battery **10** according to Embodiment 1 of the present disclosure may have a hole **340** formed therein. For example, at least one hole **340** may be formed in a portion of the second

exterior material **300**.

[0077] As described above, the gas inside the first exterior material **200** may permeate through the gas permeable first exterior material **200**. In contrast, the gas that has permeated through the gas-permeable first exterior material **200** may not permeate through the second exterior material **300**, which includes the metal layer **320**. At this time, the gas that has permeated through the first exterior material **200** may be discharged to the exterior of the secondary battery **10** through the hole **340** of the second exterior material **300**.

[0078] The hole **340** formed in the second exterior material **300** may vary in shape, size, and number. For example, the hole **340** may be formed at a portion corresponding to the portion of the first exterior material **200** that accommodates the electrode assembly **100**. In addition, multiple holes **340** may be formed.

[0079] The secondary battery **10** may have a configuration in which the first exterior material **200** seals the electrode assembly **100**, and then the second exterior material **300** encloses the first exterior material **200**. At this time, since only the periphery of the second exterior material **300** is bonded to the first exterior material **200**, a space (gap) may be formed between the first exterior material **200** and the second exterior material **300** in the unbonded portions. Gas that permeates through the first exterior material **200** may travel along the space (gap) between the first exterior material **200** and the second exterior material **300** and be discharged to the exterior through the hole **340** formed in the second exterior material **300**. Thus, the positions where the holes **340** may be formed in the second exterior material **300** may be more diverse.

[0080] In addition, during the manufacturing process of the secondary battery **10**, the second exterior material **300** may enclose the first exterior material **200** sealed in the state where the electrode assembly **100** is accommodated therein. At this time, the holes **340** may be formed in the second exterior material **300** before the second exterior material encloses the first exterior material **200**. Therefore, the degree of flexibility regarding the positions, number, and shapes of the holes **340** in the second exterior material **300** may be increased.

[0081] The secondary battery **10** according to Embodiment 1 of the present disclosure may efficiently discharge gas generated inside the secondary battery **10** to the exterior, as the second exterior material **300** has the holes **340** formed therein. Through this, the secondary battery **10** may prevent or suppress problems of functional degradation. In addition, since the secondary battery **10** according to Embodiment 1 of the present disclosure has the first exterior material **200** and the second exterior material **300** installed as described above, the occurrence of venting caused by internal gas may be delayed or prevented, thereby improving safety.

Embodiment 2

[0082] FIG. **4** is a perspective view schematically illustrating a secondary battery **10** according to Embodiment 2 of the present disclosure.

[0083] Hereinafter, detailed descriptions of configurations identical to those of the secondary battery **10** according to Embodiment 1 of the present disclosure will be omitted, and only the differences will be described.

[0084] The secondary battery **10** according to Embodiment 2 of the present disclosure may differ from the secondary battery **10** according to Embodiment 1 of the present disclosure in the form of the second exterior material **300**.

[0085] The second exterior material **300** of the secondary battery **10** according to Embodiment 2 of the present disclosure may include multiple layers. For example, the second exterior material **300** may include a sealant layer **310** and a metal layer **320**.

[0086] The sealant layer **310** of the second exterior material **300** may be thermally sealable. For example, the sealant layer **310** may include a thermally sealable material.

[0087] The metal layer **320** of the second exterior material **300** may include a metallic material. For example, the metal layer **320** of the second exterior material **300** may include stainless steel (SUS). In addition, the metal layer **320** may have its outer surface coated with an insulating material. As an

example, the outer surface of the metal layer **320** may be coated with nylon. Here, the outer surface of the metal layer **320** may refer to the surface opposite to the surface facing the sealant layer **310**. [0088] Referring to FIG. 4, in the secondary battery **10** according to Embodiment 2 of the present disclosure, the opposing sealant layers **310** of the second exterior material **300** may be bonded to seal the first exterior material **200**. For example, the opposing sealant layers **310** of the second exterior material **300** may be bonded through sealing to ensure that the first exterior material **200** is sealed. For example, the first exterior material **200** may seal the electrode assembly **100** accommodated therein, and the second exterior material **300** may seal the first exterior material **200** therein.

[0089] As an example, the second exterior material **300** may cover the first exterior material **200**, which accommodates the electrode assembly **100**, from both sides. At this time, the peripheral portion of the second exterior material **300** may be sealed through thermal sealing. Here, the sealant layer **310** disposed at the peripheral portion of the second exterior material **300** may be sealed.

[0090] In the secondary battery **10** according to Embodiment 2 of the present disclosure, the second exterior material **300** may not be bonded to the first exterior material **200**, and the opposing portions of the second exterior material **300** may be bonded to each other. Accordingly, the first exterior material **200** may be accommodated inside the second exterior material **300** in the state where it is invisible from the outside (see, e.g., FIG. 4).

[0091] Since the first exterior material **200** is sealed in the state where it is accommodated inside the second exterior material **300**, even when a crack occurs in the first exterior material **200**, for example, the electrolytic solution will not leak out of the second exterior material **300**. Thus, the stability of the secondary battery **10** may be improved.

[0092] Referring to FIG. 4, the second exterior material **300** of the secondary battery **10** according to Embodiment 2 of the present disclosure may have holes **340** with a different shape compared to Embodiment 1. However, this is merely an example, and the shape, position, and number of the holes **340** may vary.

[0093] In addition to the forms of the second exterior material **300** described in various embodiments of the present disclosure, the form of the second exterior material **300** may vary to achieve structural stability and efficient gas discharge.

Embodiment 3

[0094] FIG. 5 is a flowchart schematically illustrating a secondary battery manufacturing method according to Embodiment 3 of the present disclosure. FIG. 6 is a plan view schematically illustrating a state where the first exterior material **200** encloses the electrode assembly **100** in the secondary battery manufacturing method according to Embodiment 3 of the present disclosure, and FIG. 7 is a plan view schematically illustrating a state where a gas collection portion **210** is removed in the secondary battery manufacturing method according to Embodiment 3 of the present disclosure.

[0095] Referring to FIGS. 5 and 6, the secondary battery **10** manufacturing method according to Embodiment 3 of the present disclosure may include a step of sealing the first exterior material **200** that accommodates the electrode assembly **100** (S1). Here, the first exterior material **200** may refer to an exterior material that includes a gas collection portion **210** and a cup portion **220**. For example, the first exterior material **200** described in Embodiment 3 of the present disclosure may differ in form from the first exterior material **200** described in the preceding embodiments in that it includes the gas collection portion **210**. However, if necessary, it may also be in the same form as the first exterior material **200** described in the preceding embodiments. The cup portion **220** may refer to a space in which the electrode assembly **100** is disposed, and the gas collection portion **210** may be provided to be spaced apart from the cup portion **220**.

[0096] Referring to FIG. 6, the first exterior material **200** may be sealed in the state where the electrode assembly **100** is accommodated in the cup portion **220**. For example, the cup portion **220**

may accommodate the electrode assembly **100** and an electrolytic solution.

[0097] Next, a step of activating the electrode assembly **100** disposed inside the sealed first exterior material **200** by charging and discharging (S2) may be performed. Through this, electrical properties may be imparted to the electrode assembly **100**. In the secondary battery manufacturing method according to Embodiment 3 of the present disclosure, the electrode assembly **100** may be activated in a state sealed only with the first exterior material **200**. For example, in the secondary battery manufacturing method according to Embodiment 3 of the present disclosure, after the electrode assembly **100** is sealed with the first exterior material **200**, the activation may proceed while the second exterior material **300** is applied.

[0098] Meanwhile, during the activation of the electrode assembly **100**, gas may be generated inside the first exterior material **200**. The gas generated during the activation may be collected in the gas collection portion **210** of the first exterior material **200**. Accordingly, a step of discharging the gas collected in the gas collection portion **210** of the first exterior material **200** through activation (S3) may be performed. For example, in the secondary battery manufacturing method according to Embodiment 3 of the present disclosure, after the activation is performed with only the first exterior material **200** enclosing the electrode assembly **100**, the gas collected in the gas collection portion **210** may be removed or discharged.

[0099] The method for discharging the gas collected in the gas collection portion **210** may vary. For example, the gas may be discharged by forming one or more holes in the gas collection portion **210**.

[0100] After the gas collected in the gas collection portion **210** is discharged, a step of sealing the area between the gas collection portion **210** and the cup portion **220** (S4) may be performed. For example, the area between a tear line N formed in the gas collection portion **210** and the cup portion **220** may be sealed. In another example, the first exterior material may be sealed along the tear line N.

[0101] Subsequently, a step of removing the gas collection portion **210** of the first exterior material **200** (S5) may be performed. For example, in the secondary battery manufacturing method according to Embodiment 3 of the present disclosure, a portion of the first exterior material **200** may be removed.

[0102] The first exterior material **200** may be thermally sealable and gas-permeable. For example, the first exterior material may be made of a polyolefin resin such as polypropylene (PP) or polyethylene (PE). While a pouch film including multiple layers is to be removed in conventional secondary battery manufacturing processes, in the secondary battery manufacturing method according to Embodiment 3 of the present disclosure, only a portion of the relatively inexpensive first exterior material **200** is removed, thereby reducing the costs associated with removing the gas collection portion **210**. As a result, the economic efficiency of the secondary battery manufacturing process may be improved.

[0103] In the second battery manufacturing method according to Embodiment 3 of the present disclosure, the first exterior material **200** may have an arbitrary tear line N formed between the gas collection portion **210** and the cup portion **220** (see, e.g., FIG. 6). Through cutting along the tear line N, the gas collection portion **210** may be removed. After the gas collection portion **210** is removed, the first exterior material **200** may remain with only the cup portion **220** containing the electrode assembly **100** (see, e.g., FIG. 7).

[0104] FIG. 8 is a plan view schematically illustrating a state where the second exterior material **300** encloses the first exterior material **200** in the secondary battery manufacturing method according to Embodiment 3 of the present disclosure.

[0105] After the gas collection portion **210** is removed, a step of enclosing the first exterior material **200** with the second exterior material **300** (S7) may be performed. For example, by enclosing the first exterior material **200** with the second exterior material **300**, the electrode assembly **100** may be sealed again, for example, in double layers. For example, the second exterior

material **300** may be coupled to the first exterior material **200** to reseal the electrode assembly **100**. Here, the first exterior material **200** and the second exterior material **300** may be bonded to each other through thermal sealing. In the secondary battery manufacturing method according to Embodiment 3 of the present disclosure, the first exterior material **200** and the second exterior material **300** may be partially bonded. Thus, the first exterior material **200** and the second exterior material **300** may have at least some unbonded portions.

[0106] In the secondary battery manufacturing method according to Embodiment 3 of the present disclosure, activation and gas removal may be performed while the first exterior material **200** accommodates the electrode assembly **100** and the electrolytic solution. Accordingly, the possibility of sealing defects caused by the electrolytic solution during the sealing process of the second exterior material **300** may be reduced. As a result, the possibility of insulation defects caused by sealing defects may also be reduced. In addition, the sealing of the second exterior material **300** may be facilitated, thereby improving the efficiency of the secondary battery manufacturing process.

[0107] In order to efficiently discharge gas generated during the use of the completed secondary battery **10**, the secondary battery manufacturing method according to Embodiment 3 of the present disclosure may further include a step of forming at least one hole **340** in a portion of the second exterior material **300** (S6).

[0108] For example, the step of forming the hole **340** in the second exterior material **300** (S6) may be performed before the step of enclosing the first exterior material **200** with the second exterior material **300** (S7). Since the hole **340** in the second exterior material **300** is formed before the sealing of the second exterior material **300** in the secondary battery manufacturing method according to Embodiment 3 of the present disclosure, the degree of flexibility in forming the hole **340** may be increased. Meanwhile, in another embodiment, the hole **340** in the second exterior material **300** may also be formed after the step of enclosing the first exterior material **200** with the second exterior material **300** (S7).

[0109] Referring to FIG. 9, multiple secondary batteries **10** according to an embodiment of the present disclosure may be coupled to form a battery module or a battery pack **600**. The battery pack **600** may be included in a vehicle **700**, such as an electric vehicle (EV) or a hybrid vehicle (HV). Here, the secondary battery **10** described above may be applied to the secondary batteries **10**. The battery pack **600** may supply power to the motor through an inverter provided in the vehicle **700**, thereby driving the vehicle **700**.

[0110] The battery pack **600** may include a battery management system (not illustrated). For example, the vehicle **700** may include a battery management system. In this case, the battery management system may be an onboard system included in the vehicle **700**.

[0111] While the present disclosure has been described with limited embodiments and drawings, the present disclosure is not limited thereto, and various embodiments may be made by one of ordinary skill in the art within the scope equivalent to the technical idea of the present disclosure and the claims described set forth below.

Claims

1. A secondary battery comprising: an electrode assembly; a first exterior material enclosing the electrode assembly such that the electrode assembly is sealed; a second exterior material enclosing an outer surface of the first exterior material, wherein at least a portion of the second exterior material is unbonded to the first exterior material; and an electrode lead electrically connected to the electrode assembly and provided to extend from an interior to an exterior of the first exterior material and the second exterior material.

2. The secondary battery of claim 1, wherein the first exterior material has thermal sealability and gas permeability.

3. The secondary battery of claim 1, further comprising: an electrolyte accommodated within the interior of the first exterior material along with the electrode assembly.
 4. The secondary battery of claim 1, wherein the second exterior material has thermal sealability and includes a sealant layer provided to face the first exterior material.
 5. The secondary battery of claim 4, wherein at least a portion of the sealant layer is bonded to the first exterior material by sealing.
 6. The secondary battery of claim 4, wherein the second exterior material is bonded to the sealant layer provided to face the first exterior material by sealing such that the first exterior material is sealed.
 7. The secondary battery of claim 1, wherein the second exterior material includes: a metal layer including a metallic material; and an insulating layer including an insulating material.
 8. The secondary battery of claim 1, wherein the second exterior material includes: a metal layer including stainless steel (SUS), and an outer surface of the metal layer is coated with nylon.
 9. The secondary battery of claim 1, wherein a portion of the second exterior material includes at least one hole formed therein.
 10. A method of manufacturing a secondary battery, the method comprising: sealing a first exterior material accommodating an electrode assembly in a cup portion; charging and discharging the electrode assembly provided in the sealed first exterior material to activate the electrode assembly; discharging gas collected in a gas collection portion of the first exterior material through the activation; removing the gas collection portion of the first exterior material; and enclosing the first exterior material with a second exterior material.
 11. The method of claim 10, further comprising: sealing between the cup portion and the gas collection portion after the discharging the collected gas.
 12. The method of claim 10, further comprising: forming at least one hole in a portion of the second exterior material before the enclosing the first exterior material.
 13. The method of claim 10, wherein, in the enclosing the first exterior material, the first exterior material and the second exterior material are bonded to each other by thermal sealing.
 14. A battery pack comprising the secondary battery of claim 1.
 15. A vehicle comprising the battery pack of claim 14.
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